METRIC

MIL-DTL-51027C
16 December 2021
SUPERSEDING
MIL-DTL-51027B
12 February 2010

DETAIL SPECIFICATION

ANTISETTING COMPOUND, DECONTAMINATING SLURRY, M2

Inactive for new design after 24 November 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of antisetting compound.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications. The following specification forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of this document are those cited in the solicitation or contract.

FEDERAL SPECIFICATION

O-S-635 - Sodium Polyphosphates, Technical

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to DLA Aviation, VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.dla.mil

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

A-A-59282 - Chemicals, Analytical; General Specification for

(Copies of this document are available online at https://quickserach.dla.mil.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM C25	-	Standard Test Methods for Chemical Analysis of Limestone, Quicklime, and Hydrated Lime
ASTM D1193	-	Standard Specification for Reagent Water
ASTM E11	-	Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves

(Copies of these documents are available online at http://www.astm.org/)

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

USP-NF - U.S. Pharmacopeia-National Formulary

(Copies of this document are available online at http://www.usp.org/)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Antisetting compound shall be manufactured from the following materials:

a. Citric acid, anhydrous, fine granular, conforming to the USP-NF.
b. Sodium tripolyphosphate, technical grade, conforming to type III of O-S-635.
c. Calcium oxide, technical grade, 90 percent pure minimum, 94 percent nonvolatile matter minimum, 95 percent minimum particle size retained on No. 7 sieve.

3.2 Chemical composition. Antisetting compound shall be a thoroughly blended mixture conforming to the characteristics of table I when tested as specified herein.

TABLE I. Chemical composition.

Material	Percent by weight	Test paragraph
Citric acid, anhydrous	46.5 ±1.0	4.3.1.1
Sodium tripolyphosphate	50.5 ±1.0	4.3.1.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

Calcium oxide	3.0 ±0.2	4.3.1.3

3.3 Particle size. A minimum of 95 percent by weight of the antisetting compound shall pass through a No. 4 sieve when tested as specified in 4.3.4.

3.4 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirement specified herein is classified as conformance inspection (see 4.2).

4.2 Conformance inspection.

4.2.1 Lotting. A lot shall consist of the antisetting compound produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3)

4.2.2 Sampling.

4.2.2.1 For antisetting compound test. Sampling for test shall be conducted in accordance with table II. A representative specimen of approximately 0.5 kilograms (kg) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for test.

Number of containers in batch or lot	Number of sample containers
2 - 25	2
26 - 150	3
151 - 1,200	5
1,201 - 7,000	8
7,001 - 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For test. Approximately equal portions of all of the specimens except one taken in 4.2.2.1 shall be thoroughly mixed to form a composited specimen of no less than 0.5 kg. The composited specimen and the remaining individual specimen shall be tested separately and in duplicate as specified in 4.3, and the duplicates averaged. Failure of the average of any test by either the composited or the individual specimen to meet specified requirements shall be cause for rejection of the lot represented.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

4.3 Tests. Water in accordance with ASTM D1193 and chemicals in accordance with A-A-59282 shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.3.1 Weight Percent Determination of components within anti-setting compound.

4.3.1.1 Citric Acid. Weight percent citric acid shall be determined by peak area comparison using liquid or ion chromatography. Calibration shall be done in accordance with the recommendations of the equipment manufacturer but shall include at a minimum comparison to known standards of all ions of interest at concentrations above, below, and near the target value as well as a blank standard. Regular checks of the calibration shall be performed in accordance with the equipment manufacturer's recommendations. Calcium oxide may be precipitated out prior to analysis provided a blank determination is run to correct any affect this has on the reading.

4.3.1.2 Sodium Tripolyphosphate. Weight percent sodium tripolyphosphate shall be determined by peak area comparison using ion chromatography. Calibration shall be done in accordance with the recommendations of the equipment manufacturer but shall include at a minimum comparison to known standards of all ions of interest at concentrations above, below, and near the target value as well as a blank standard. Regular checks of the calibration shall be performed in accordance with the equipment manufacturer's recommendations. Calcium oxide may be precipitated out prior to analysis provided a blank determination is run to correct any affect this has on the reading. Note:Tripolyphosphate can hydrolysis into other compounds. Steps may need to be taken to prevent this and at a minimum, all tests shall be run at approximately the same time from dissolution.

4.3.1.3 Calcium oxide. Weight percent calcium oxide shall be determined in accordance with ASTM C25 using either method for determination of calcium oxide content.

4.3.2 Granulation. Place an accurately weighed specimen of 100.0 g on a No. 4 sieve as identified in ASTM E11 (see 3.3) and assemble with a bottom pan. Cover and shake for three minutes by hand or by means of a mechanical shaker geared to produce 300 ±15 gyrations and 150 ±10 taps of a striker per minute. Weigh the portion passed by the sieve and calculate the percent passing.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Antisetting compound, decontaminating slurry, M2 is intended for use in retarding settling and setting in decontaminating slurry mixtures.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Packaging requirements (see 5.1).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, observed or calculated values should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method in ASTM E29, "Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications".

6.5 Subject term (key word) listing.

Calcium oxide
Citric acid
Decontamination
Mixture
Sodium tripolyphosphate
Liquid chromatography
Ion chromatography

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - EA
Navy - MC
DLA - GS

Preparing Activity:
DLA - GS3

(Project 6850-2022-002)

Review Activities:
Army - MD
Navy - YD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST database at https://assist.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.